CURRICULUM VITAE

Kevin J. Anchukaitis

Address: University of Arizona
Laboratory of Tree-Ring Research
Bryant Bannister Tree-Ring Building
1215 E. Lowell Street, Tucson, AZ 85721-0045
orcid: 0000-0002-8509-8080
email: kanchukaitis@arizona.edu
github: https://github.com/kanchukaitis
internet: https://kanchukaitis.github.io/

Appointments

University of Arizona — Tucson, AZ

2025 – Director, Laboratory of Tree-Ring Research
Professor of Dendrochronology
Joint Appointment, Department of Geosciences
Associate, Center for Climate Adaptation Science and Solutions (CCASS)
Affiliated Faculty, Graduate Interdisciplinary Program in Global Change
Affiliated Faculty, Latin American Studies

2021 – Professor, School of Geography, Development, and Environment
2017 – **2025** Chair, Graduate Interdisciplinary Program in Global Change
2015 – **2021** Associate Professor, School of Geography, Development, and Environment

Lamont Doherty Earth Observatory of Columbia University — New York, NY

2012 – Adjunct Senior Research Scientist

Woods Hole Oceanographic Institution — Woods Hole, MA

2012 – **2015** Assistant Scientist

Lamont Doherty Earth Observatory of Columbia University — New York, NY

2010 – **2012** Lamont Assistant Research Professor
2009 – **2010** Doherty Associate Research Scientist
2007 – **2009** Lamont-Doherty Postdoctoral Research Fellow

Education

University of Arizona — Tucson, AZ

2007 Ph.D., Geosciences

University of Tennessee — Knoxville, TN

2002 M.S., Geography

Georgetown University — Washington, DC

1998 B.S., (*cum Laude*) Science, Technology, & International Affairs
Major Field: Environmental Science (with Honors)
Foreign study at University of Costa Rica, Institute for Tropical Studies (Golfito), 1996 – 1997

Publications

Submitted or in Revision († = postdoctoral author, ‡ = student author)

Allen, K., M. Freund, **K.J. Anchukaitis**, J. Palmer, N. English, R. Evans, A. Adriaanse-Tucker, J. King, E.R. Cook, B.M. Buckley, P. Krusic, P. Baker, Common Era temperature variability for two seasons in Tasmania, southern Australia, in revision, 2026

Zhang, X., J. Li, S-P Xie, E.R. Cook, R.D'Arrigo, **K.J. Anchukaitis**, Unprecedented 21st-century water scarcity in southwestern North America since 500 CE, submitted, 2026

Anchukaitis, K.J., J.O. Kaplan, B.I. Cook, D. Penny, Revisiting climate variability and the collapse of Greater Angkor, in press, 2026

Peer-Reviewed and Invited Papers († = postdoctoral author, ‡ = student author)

[155] Nguyen-Xuan Hau, M. Sano, T. Nakatsuka, **K.J. Anchukaitis**, Thanh Ngo-Duc, Ying-Shuo Chang, Shin-Hao Chen, Ssu-Hao Wang, Hsin-Lin Wei, Zhen Li, Le-Thi Thuong, Tran Binh Minh, I-Ching Chen, ENSO-modulated upstream rainout and convection dominate precipitation oxygen isotopes recorded in tree rings from Taiwan's cloud forests, *Climate Dynamics*, accepted, 2026

[154] ‡ Kunz, M., M.C.A. Torbenson, F. Reinig, E. Martinez del Castillo, U. Büntgen, R. Wilson, I.K. Homfeld, G. King, E. Reid, **K.J. Anchukaitis**, V. Trouet, K.E. King, G.L. Harley, J.T. Maxwell, A. Csank, E. Broadman, E. Kuhl, J. Edwards, P. Römer, B. Günther, C. Gerber, J. Esper, Exorcizing divergence in tree-ring density along the Rocky Mountains, *Journal of Geophysical Research – Biogeosciences*, 131, 5, doi:10.1029/2026JG009783, 2026

[153] ‡ González-Mendez, I., **K.J. Anchukaitis**, D. Pons, K. Morino, T.G. Anderson, S. Belmecheri, L. Andreu-Hayles, Oxygen isotopes in tree rings track neotropical climate dynamics, *Geophysical Research Letters*, 53, 4, doi:10.1029/2025GL120744, 2026

[152] † Wang, F., E.K. Wise, **K.J. Anchukaitis**, Q. Chang, M.P. Dannenberg, Evaluation of daily gridded climate products using in situ FLUXNET data and ecological modeling, *Environmental Research Letters*, 21, 2, 024023, doi:10.1088/1748-9326/ae384b, 2026

[151] ‡ Anderson, T.G., D. Pons, Z. Guido, L. Tun, **K.J. Anchukaitis**, Usability of Seasonal Climate Forecasts in the Guatemalan Dry Corridor, *Weather, Climate, and Society*, 18(1), 127–147, doi:10.1175/WCAS-D-25-0001.1, 2026

[150] ‡ González-Mendez, I., D. Pons, T.G. Anderson, I. Ayes-Rivera, **K.J. Anchukaitis**, Impacts of Spatial and Temporal Station Availability on Gridded Precipitation Products in Central America, *Earth and Space Science*, 12(12), doi:10.1029/2025EA004720, 2025

[149] ‡ Edwards, J., **K.J. Anchukaitis**, F. Molina, L. Andreu-Hayles, R. D'Arrigo, G. von Arx, Resolution and frequency-dependent climate signals in an Arctic tree-ring temperature reconstruction of the last millennium, *Geophysical Research Letters*, 52, 20, e2025GL117846, doi:10.1029/2025GL117846, 2025

[148] ‡ Saunders, A., B. Tellman, E. Benami, **K.J. Anchukaitis**, S. Hossain, A. Bennett, A.K.M. Saiful Islam, J. Giezendanner, Sensitivity to Data Choice for Index-Based Flood Insurance, *Earth's Future*, 13, 9, e2025EF005966, doi:10.1029/2025EF005966, 2025

[147] ‡ Hwangbo, N., K.A. McKinnon, **K.J. Anchukaitis**, Simultaneous optimal target season estimation and local climate reconstruction using tree rings, *Geophysical Research Letters*, 52, 17, e2025GL116495, doi:10.1029/2025GL116495, 2025

[146] ‡ Verkerk, M., T.J. Aubry, C. Smith, P.O. Hopcroft, M. Sigl, J.E. Tierney, **K.J. Anchukaitis**, M. Osman, A. Schmidt, M. Toohey, Using reduced-complexity volcanic aerosol and climate models to produce large ensemble simulations of Holocene temperature, *Climate of the Past*, 21, 10, 1755–1778, doi:10.5194/cp-21-1755-2025, 2025

[145] ‡ Anderson, T.G., D. Pons, M.J. Taylor, A. Xuruc, H. Rodríguez-Salvatierra, Z. Guido, J.A. Sullivan, D. Liverman, **K.J. Anchukaitis**, Complexity and mediating factors in farmers' climate perceptions and agricultural adaptation strategies in the Guatemalan Dry Corridor, *Climatic Change*, 178, 139, doi:10.1007/s10584-025-03978-5, 2025

[144] Tierney, J.E., E.J. Judd, M.B. Osman, J.M. King, O.J. Truax, N. Steiger, D. Amrhein, **K.J. Anchukaitis**, Advances in paleoclimate data assimilation, *Annual Review of Earth and Planetary Sciences*, 53, 625-650, doi:10.1146/annurev-earth-032320-064209, 2025

[143] ‡ Harris, K.D., J.E. Tierney, S. Munoz, **K.J. Anchukaitis**, Climate variability in the Gulf of California over the last 1300 years, *Paleoceanography and Paleoclimatology*, 40, 2, e2024PA004912, doi:10.1029/2024PA004912, 2025

[142] † Leland, C., N. Davi, R. D'Arrigo, L. Andreu-Hayles, A. Pacheco-Solana, J. Edwards, **K.J. Anchukaitis**, T. Porter, M.F.J. Pisaric, M. Mant, T. Galloway, and R. Oelkers, Tree-ring Evidence of the Elusive 1959 Summer Cold Event in Northwestern North America, *Arctic, Antarctic, and Alpine Research*, 57(1), doi:10.1080/15230430.2024.2445945, 2025

[141] Marshall, L.R., A. Schmidt, A.P. Schurer, N.L. Abraham, L.J Lucke, R. Wilson, **K.J. Anchukaitis**, G. Hegerl, B. Johnson, B.L. Otto-Bliesner, E.C. Brady, M. Khodri, K. Yoshida, Last Millennium volcanic forcing and climate response using SO_2 emissions, *Climate of the Past*, 21, 161–184, doi:10.5194/cp-21-161-2025, 2025

[140] ‡ González-Mendez, I., **K.J. Anchukaitis**, C. Pearson, D. Pons, D. Frank, J. Edwards, E. Luna, W. Pérez, Annual chronology and climate signals in *Swietenia macrophylla* and *Cedrela odorata* (Meliaceae) in the Maya Lowlands, *Paleoceanography and Paleoclimatology*, 40, 1, e2024PA005036, doi:10.1029/2024PA005036, 2025

[139] ‡ Edwards, J., W.L. Tintor, A.F. Nolin, C.A. Woodhouse G. von Arx, **K.J. Anchukaitis**, Multiple elevation-dependent climate signals from quantitative wood anatomical measurements of Rocky Mountain bristlecone pine, *Journal of Geophysical Research – Biogeosciences*, 130, 1, e2024JG008307, doi: 10.1029/2024JG008307, 2025

[138] Williams, A.P., K.A. McKinnon, **K.J. Anchukaitis**, A. Gershunov, A.M. Varuolo-Clarke, R.E.S Clemesha, H. Liu, Anthropogenic intensification of cool-season precipitation is not yet detectable across the western United States, *Journal of Geophysical Research – Atmospheres*, 129, 12, e2023JD040537, doi:10.1029/2023JD040537, 2024

[137] Cook, B.I., E.R. Cook, **K.J. Anchukaitis**, D. Singh, Characterizing the 2010 Russian heatwave-Pakistan flood concurrent extreme over the last millennium using the Great Eurasian Drought Atlas, *Journal of Climate*, 37, 17, 4389–4401, doi:10.1175/JCLI-D-23-0773.1, 2024

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[135] **Anchukaitis, K.J.**, R Touchan, D.M. Meko, D. Kherchouche, S. Slimani, F. Sivrikaya, R. Ilmen, I. Mitsopoulos, J. Stephan, J. Attieh, F. Hasnaoui, J.J. Camarero, R. Sánchez-Salguero, F. Guibal, A. Piermattei, A. Christou, J. Krcmaric, B.I. Cook, Enhancing spatiotemporal reconstructions of hydroclimate variability across the Mediterranean during the last millennium, *Climate Dynamics*, 62, 5351–5371, doi:10.1007/s00382-024-07166-6, 2024

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[132] Herrera, D.A., B.I Cook, J. Fasullo, **K.J. Anchukaitis**, M.J. Alessi, C.J. Martinez, C.P. Evans, X. Li, K.N. Ellis, R. Mendez-Tejeda, T.R. Ault, A. Centella-Artola, T.S. Stephenson, M.A. Taylor, Observed changes in hydroclimate attributed to human forcing, *PLOS Climate*, 2(11): e0000303, doi:10.1371/journal.pclm.0000303, 2023

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[130] Burke, A., H. Innes, L. Crick, **K.J. Anchukaitis**, M. Byrne, W. Hutchison, J. McConnell, K. Moore, J. Rae, M. Sigl, and R. Wilson, High sensitivity of summer temperatures to stratospheric sulfur loading from volcanoes in the Northern Hemisphere, *Proceedings of the National Academy of Sciences*, 120, 47, e2221810120, doi:10.1073/pnas.2221810120, 2023

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[128] Zhu, F., J. Emile-Geay, **K.J. Anchukaitis**, N. McKay, S. Stevenson, Z. Meng, A pseudoproxy emulation of the PAGES 2k database using a hierarchy of proxy system models, *Scientific Data*, 10, 624, doi: 10.1038/s41597-023-02489-1, 2023

[127] Schneider, L., O. Konter, J. Esper, **K.J. Anchukaitis**, Constraining the 19th century temperature baseline for global warming, *Journal of Climate*, 36, 18, 6261–6272, doi:10.1175/JCLI-D-22-0806.1, 2023

[126] ‡ Dye, L.A., B.L. Coulthard, B. Hatchett, I.K. Homfeld, T.N. Salazar, J.S. Littell, **K.J. Anchukaitis**, The severity of the 2014-2015 snow drought in the Oregon Cascades in a multicentury context, *Water Resources Research*, 59, 5, doi:10.1029/2022WR032875, 2023

[125] ‡ King, J., **K.J. Anchukaitis**, K. Allen, T. Vance, A. Hessl, Trends and variability in the Southern Annular Mode during the Common Era, *Nature Communications*, 14, 2324, doi:10.1038/s41467-023-37643-1, 2023

[124] † Leland, C., R. D'Arrigo, N. Davi, **K.J. Anchukaitis**, L. Andreu-Hayles, T.J. Porter, T. Galloway, M. Mant, G. Wiles, R. Wilson, S. Beaulieu, R. Oelkers, B. Gaglioti, M.P. Rao, E. Reid, T. Nixon, A Spatiotemporal Assessment of Extreme Cold in Northwestern North America Following the Unidentified 1809 CE Volcanic Eruption, *Paleoceanography and Paleoclimatology*, 38, 5, doi:10.1029/2022PA004581, 2023

[123] ‡ Heeter, K., G. Harley, J. Abatzoglou, **K.J. Anchukaitis**, E.R. Cook, B.L Coulthard, L.A. Dye, and I.K. Homfled, Unprecedented 21st century heat across the Pacific Northwest of North America, *npj Climate and Atmospheric Science*, 6, 5, doi:10.1038/s41612-023-00340-3, 2023

[122] ‡ Walker, M., A. Mueller, K. Allen, P. Fenwick, V. Agrawal, **K.J. Anchukaitis**, A. Hessl, High Resolution Radiocarbon Spike Confirms Tree Ring Dating with Low Sample Depth, *Dendrochronologia*, 77, 126048, doi:10.1016/j.dendro.2022.126048, 2022

[121] † Leland, C., L. Andreu-Hayles, E.R. Cook, **K.J. Anchukaitis**, O. Byambasuren, N. Davi, A. Hessl, D. Martin-Benito, B. Nachin, N. Pederson, Impacts of Climate and Tree Morphology on Tree-ring Stable Isotopes in Central Mongolia, *Tree Physiology*, 43, 4, 539–555, doi:10.1093/treephys/tpac142, 2022

[120] Cook, B.I. J.E. Smerdon, E.R. Cook, A.P. Williams, **K.J. Anchukaitis**, J.S. Mankin, K. Allen, L. Andreu-Hayles, T.R Ault, S. Belmecheri, S. Coats, B. Coulthard, B. Fosu, P. Grierson, D. Griffin, D. Herrara, M. Ionita, F. Lehner, C. Leland, K.D. Marvel, M. Morales, V. Mishra, J. Ngoma, H. Nguyen, A. O'Donnell, J. Palmer, M.P. Rao, M. Rodriguez, D. Stahle, S. Stevenson, U.K. Thapa, A. Varuolo-Clarke, E. Wise, Megadroughts in the Common Era and the Anthropocene, *Nature Reviews Earth and Environment*, 3, 741–757, doi:10.1038/s43017-022-00329-1, 2022

[119] ‡ Grzywacz, Z., A. Hessl, **K.J. Anchukaitis**, S. Sharma, M.N. Evans, S. Nichols, Investigating the potential for Southern Hemisphere climate reconstruction using stable isotopes in Tasmanian tree rings, *Dendrochronologia*, 76, 126016, doi:10.1016/j.dendro.2022.126016, 2022

[118] Degroot, D., **K.J. Anchukaitis**, J.E. Tierney, F. Riede, A. Manica, E. Moesswilde, N. Gauthier, The History of Climate and Society: A Review of the Influence of Climate Change on the Human Past, *Environmental Research Letters*, 17, 10, 103001, doi:10.1088/1748-9326/ac8faa, 2022

[117] **Anchukaitis, K.J.** and J.E. Smerdon, Progress and uncertainties in global and hemispheric temperature reconstructions of the Common Era, *Quaternary Science Reviews*, 286, 107537, doi:10.1016/j.quascirev.2022.107537, 2022

[116] ‡ Reed, E.V, D.M Thompson, **K.J. Anchukaitis**, Coral-based Sea Surface Salinity Reconstructions and the Role of Observational Uncertainties in Inferred Variability and Trends, *Paleoceanography and Paleoclimatology*, 37, 6, doi:10.1029/2021PA004371, 2022

[115] ‡ Trinies, C.A. A.G. Bunn, C. Robertson, **K.J. Anchukaitis**, Dendroclimatology of yellow-cedar (*Callitropsis nootkatensis*) and temperature variability on the western slopes of the North Cascades in Washington State, USA from 1333 to 2015 CE, *Tree-Ring Research*, 78(2):113-128, doi:10.3959/2021-20, 2022

[114] Mackay, H., G. Plunkett, B. Jensen, T. Aubry, C. Corona, W.M. Kim, M. Toohey, M. Sigl, M. Stoffel, **K.J. Anchukaitis**, C. Raible, M. Bolton, J. Manning, T. Newfield, N. Di Cosmo, F. Ludlow, C. Kostick, Z. Yang, L. Coyle McClung, M. Amesbury, A. Monteath, P. Hughes, P. Langdon, D. Charman, R. Booth, K. Davies, A. Blundell, G. and Swindles, The 852/3 CE Mount Churchill eruption: examining the potential climatic and societal impacts and the timing of the Medieval Climate Anomaly in the North Atlantic Region, *Climate of the Past*, 18, 6, 1475–1508, doi:10.5194/cp-18-1475-2022, 2022

[113] ‡ Zhu, F., J. Emile-Geay, **K.J. Anchukaitis**, G.J. Hakim, A. Wittenberg, M. Morales, M. Toohey, J. King, A re-appraisal of the ENSO response to volcanism with paleoclimate data assimilation, *Nature Communications*, 13, 747, doi:10.1038/s41467-022-28210-1, 2022

[112] ‡ Edwards, J., **K.J. Anchukaitis**, B.E. Gunnarson, C. Pearson, K. Seftigen, G. von Arx, H.W. Linderholm, The origin of tree-ring reconstructed summer cooling in Northern Europe during the 18th century eruption of Laki, *Paleoceanography and Paleoclimatology*, 37, 2, e2021PA004386, doi:10.1029/2021PA004386, 2022

[111] † Gauthier, N., B.L. Coulthard, **K.J. Anchukaitis**, Pattern-based downscaling of snowpack variability in the western United States, *Climate Dynamics*, 58, 3225–3241, doi:10.1007/s00382-021-06094-z, 2022

[110] Brice, R., B.L. Coulthard, I.K. Homfeld, L.A. Dye, **K.J. Anchukaitis**, Paleohydrological context for recent floods and droughts in the Fraser River basin, *Environmental Research Letters*, 16, 124074, doi:10.1088/1748-9326/ac3daf, 2021

[109] Shishov, V., I. Tychkov, **K.J. Anchukaitis**, G.K. Zelenov, E.A. Vaganov, A band model of cambium development: opportunities and prospects, *Forests*, 12(10):1361, doi:10.3390/f12101361, 2021

[108] ‡ Windler, G., J.E. Tierney, **K.J. Anchukaitis**, Glacial-interglacial shifts dominate tropical Indo-Pacific hydroclimate during the late Pleistocene, *Geophysical Research Letters*, 48, 15, e2021GL093339, doi:10.1029/2021GL093339, 2021

[107] ‡ Anderson, T.G., D.A. Christie, R.O. Chávez, M. Olea, **K.J. Anchukaitis**, Spatiotemporal peatland productivity and climate relationships across the western South American Altiplano, *Journal of Geophysical Research - Biogeosciences*, 126, 6, e2020JG005994, doi:10.1029/2020JG005994, 2021

[106] ‡ King, J.M., **K.J. Anchukaitis**, J.E. Tierney, G.J. Hakim, J. Emile-Geay, F. Zhu, R. Wilson, A data assimilation approach to last millennium temperature field reconstruction using a limited high-sensitivity proxy network, *Journal of Climate*, 34(17): 7091–7111, doi:10.1175/JCLI-D-20-0661.1, 2021.

[105] Buntgen, U., K. Allen, **K.J. Anchukaitis**, D. Arseneault, E. Boucher, A. Brauning, S. Chatterjee, P. Cherubini, O.V. Churakova, C. Corona, F. Gennaretti, J. Grießinger, S. Guillet, J. Guiot, B. Gunnarson, S. Helama, P. Hochreuther, M.K. Hughes, P. Huybers, A.V. Kirdyanov, P.J. Krusic, J. Ludescher, W. J.-H. Meier, V.S. Myglan, K. Nicolussi, C. Oppenheimer, F. Reinig, M.W. Salzer, K. Seftigen, A.R. Stine, M. Stoffel, S. St. George, E. Tejedor, A. Trevino, V. Trouet, J. Wang, R. Wilson, B. Yang, G. Xu, J. Esper, The influence of decision-making in tree ring-based climate reconstructions, *Nature Communications*, 12, 3411, doi:10.1038/s41467-021-23627-6, 2021

[104] Degroot, D., **K.J. Anchukaitis**, M. Bauch, J. Burnham, F. Carnegy, J. Cui, K. de Luna, P. Guzowski, G. Hambrecht, H. Huhtamaa, A. Izdebski, K. Kleemann, E. Moesswilde, T. Newfield, Q. Pei, E. Xoplaki, N. Zappia. Towards a Rigorous Understanding of Societal Responses to Climate Change, *Nature*, 591, 539-550, doi:10.1038/s41586-021-03190-2, 2021

[103] Coulthard, B.L., **K.J. Anchukaitis**, G.T. Pederson, E.R. Cook, J. Littell, D. Smith, Snowpack signals in North American tree rings, *Environmental Research Letters*, 16, 034037, doi:10.1088/1748-9326/abd5de, 2021

[102] Williams, A.P., **K.J. Anchukaitis**, C. A. Woodhouse, D. M. Meko, B. I. Cook, K. Bolles, E. R. Cook, Tree rings suggest no stable cycles in Sierra Nevada cool-season precipitation, *Water Resources Research*, 57(3), e2020WR028599, doi: 0.1029/2020WR028599, 2021

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[98] Cook, B.I., J. S. Mankin, K. Marvel, A. P. Williams, J. E. Smerdon, **K.J. Anchukaitis**, Twenty-first Century Drought Projections in the CMIP6 Forcing Scenarios, *Earth's Future*, 8, 6, doi:10.1029/2019EF001461, 2020

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[95] ‡ Zhu, F., J. Emile-Geay, G.J. Hakim, J. King, **K.J. Anchukaitis**, Resolving the differences in the simulated and reconstructed climate response to volcanism over the last millennium, *Geophysical Research Letters*, 47, 8, doi:10.1029/2019gl086908, 2020

[94] Pederson, N., C. Leland, D.A. Bishop, J.K. Pearl, **K.J. Anchukaitis**, T. Mandra, M. Hopton-Ahmed, D. Martin-Benito, A framework for determining population-level vulnerability to climate: evidence for growth hysteresis in Chamaecyparis thyoides along its contiguous latitudinal distribution, *Frontiers in Forests and Global Change*, 3, 39, doi:10.3389/ffgc.2020.00039, 2020.

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[4] Gagen, M., D. McCarroll, N.J. Loader, I. Robertson, R. Jalkanen, and **K.J. Anchukaitis**, Exorcising the 'segment length curse': summer temperature reconstruction since AD1640 using non-detrended stable carbon isotope ratios from pine trees in Finnish Lapland, *The Holocene*, 17, 435-446, doi:10.1177/0959683607077012, 2007.

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Reports and Conference Proceedings

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Hessl, A.E., N. Pederson, O. Byambasuran, **K.J. Anchukaitis**, and C. Leland, How Unusual Was the 21st Century Drought in Mongolia? Placing Recent Extremes in a 2000-Year Context, *Proceedings of the Trans-disciplinary Research Conference: Building Resilience of Mongolian Rangelands*, Ulaanbaatar, Mongolia, doi:10.25675/10217/181727, 80–86, 2015.

Cook, E.R., R.D. D'Arrigo, and **K.J. Anchukaitis**, ENSO Reconstructions from Long Tree-Ring Chronologies: Unifying the Differences? in *Workshop on ENSO reconstructions for the past 500 years*, edited by H. F. Diaz, MEDIAS-FRANCE/METEO-FRANCE/NOAA, Moorea, French Polynesia, 2008.

Mentoring

Postdoctoral

Laia Andreu-Hayles, **2009–2012**, Postdoctoral Fellow and Marie Curie Postdoctoral Fellow, Lamont Doherty Earth Observatory (co-mentor with Rosanne D'Arrigo). Currently Lamont Associate Research Professor, Lamont-Doherty Earth Observatory of Columbia University.

Dario Martin-Benito, **2012–2013**, Postdoctoral Scientist, Lamont Doherty Earth Observatory. Currently Senior Scientist (*Científico Titular*) at the Forest Research Center, Instituto Nacional de Investigación, Madrid, Spain

Daniel Griffin, **2013–2014**, WHOI Postdoctoral Scholar and NOAA/UCAR Climate and Global Change Postdoctoral Fellowship, Woods Hole Oceanographic Institution. Currently Associate Professor, University of Minnesota.

Bethany Coulthard, **2016–2018**, Postdoctoral Research Scientist, Laboratory of Tree-Ring Research and School of Geography and Development, University of Arizona. Currently Marine Projects Supervisor, Cowichan Tribes, British Columbia

Nicolas Gauthier, **2019–2021**, Postdoctoral Research Scientist, Laboratory of Tree-Ring Research and School of Geography, Development, and Environment, University of Arizona. Currently Assistant Curator (tenure track), AI for Cultural and Biological Diversity, Florida Museum of Natural History, University of Florida

Bryce Belanger, **2025–present**, NSF Postdoctoral Research Fellow, Laboratory of Tree-Ring Research, University of Arizona.

Graduate Students

Gloria Jimenez, **Ph.D. 2018**, Geosciences, University of Arizona (co-advisor with Julie Cole, 2017–2018)

Jessie Pearl, **Ph.D. 2019**, Geosciences, University of Arizona (2014–2019)

Jonathan King, **Ph.D. 2022**, Geosciences, University of Arizona (2018–2022)

Talia Anderson, **M.A. 2020**; **Ph.D. 2024**, Geography, University of Arizona (2017–2024)
Julie Edwards, **M.A. 2020**; **Ph.D. 2024**, Geography, University of Arizona (2018–2024)
Kira Harris, **M.A. 2022**, Ph.D., Geography, University of Arizona (2020–present)
Ana Isabel González Mendez, **M.A. 2024**, Ph.D., Geography, University of Arizona (2022–present)

Dissertation, Thesis Committee, or External Reviewer

Petra Breitenmoser, **2013**, external referee, Ph.D., University of Bern, Switzerland
Kristina Seftigen, **2014**, *fakultetsopponent*, Ph.D., Gothenburg University, Sweden
Brewster Malevich, **2017**, committee member, Ph.D., Geosciences, University of Arizona
Diego Pons, **2017**, external committee member, Ph.D., University of Denver
Patrick Murphy, **2018** committee member, M.S., Geography, University of Arizona
Sarah Frederick, **2018** committee member, M.S., Geography, University of Arizona
Grace Windler, **2018** committee member, M.S., Geosciences, University of Arizona
Chris Zemp, **2018**, committee member, M.S., Geography, University of Arizona
Anne Billingsley, **2019**, committee member, Ph.D., Geosciences, University of Arizona
Grace Windler, **2021**, committee member, Ph.D., Geosciences, University of Arizona
Emma Reed, **2021**, committee member, Ph.D., Geosciences, University of Arizona
Lucie Luecke, **2021**, external examiner, Ph.D., School of Geosciences, University of Edinburgh
Zack Grzywacz, **2021**, external committee member, M.S., Geography, West Virginia University
William Tintor, **2021**, committee member, Ph.D., Geography, University of Arizona
Manuel Hernandez, **2022**, external committee member, Ph.D., Geography, University of North Carolina
Aria Blumm, **2022**, committee member, M.S., Geosciences, University of Arizona
Dervla Meegan Kumar, **2022**, committee member, Ph.D., Geosciences, University of Arizona
Andrew Zimmer, **2022**, committee member, Ph.D., Geography, University of Arizona
Holly Thomas, **2023**, committee member, M.S., Geosciences, University of Arizona
Brandon Strange, **2023**, committee member, Ph.D., School of Natural Resources and the Environment, University of Arizona
Jingshu Wei, **2025**, committee member, Ph.D., School of Natural Resources and the Environment, University of Arizona
Nathan Hwangbo, **2025**, external committee member, Ph.D., Statistics & Data Science, UCLA
Kai Lepley, **2025**, committee member, Ph.D., Geography, University of Arizona
Asiya Badarunnisa Sainudeen, **2026**, committee member, Ph.D., Geosciences, University of Arizona
Mudith Weerabaddana, **2026**, committee member, Ph.D., Geosciences, University of Arizona
Joaris Hernandez-Morales, **2026**, committee member, M.S., Geosciences, University of Arizona
Talitha Neesham-McTiernan, committee member, Ph.D., Geography, University of Arizona
Alex Saunders, committee member, Ph.D., Geography, University of Arizona
Dakota Larrick, committee member, Ph.D., Anthropology, University of Arizona
Elise Arellano-Thompson, committee member, Ph.D., Geography, University of Arizona
Nesrine Rouini, committee member, Ph.D., Geography, University of Arizona

Undergraduates and Interns

Kristin Campbell, **2010**, Climate and Society Intern, Columbia University (co-mentor with Ben Cook)

Hannah Aizenman, **2011**, Google Summer of Code, City University of New York and Columbia University (co-mentor with Julien Emile-Geay and Jason Smerdon)

Diego Pons, **2011**, US State Department and LASPAU Global Change Professional Fellow

Liora Hostyk, **2012**, Lamont-Doherty Earth Observatory Summer Internship (NSF REU), Barnard College and Columbia University (co-mentor with Ben Cook and Richard Seager).

Manuel Hernandez, **2013, 2014**, WHOI Summer Student Fellow and UCAR/NCAR SOARS protégé, Texas A&M University (co-mentor with Caroline Ummenhofer).

Mariel Herzog, **2013**, Lamont-Doherty Earth Observatory Summer Internship (NSF REU), University of Colorado (co-mentor with Allegra LeGrande)

Laura Fleming, **2013, 2014–2015**, Northeastern University Co–op Semester, Visiting Student at Woods Hole Oceanographic Institution

Melissa Schwan, **2016**, University of Arizona, Department of Geosciences, Senior Honors Thesis

Andrea Lara-Garcia, **2020–2021**, University of Arizona, School of Geography, Development, and Environment, Senior Honors Thesis

Zanubia Sethuraju, **2021–2022**, University of Arizona, School of Geography, Development, and Environment, undergraduate research intern and Senior Honors Thesis

Ifeoluwa Godwin Ale, **2022–2024**, University of Arizona, College of Science, Computer Science, undergraduate research programmer

Francisca Molina, **2023–2024**, University of Arizona, College of Social and Behavior Sciences, Environmental Studies and Law, undergraduate research assistant

Natalianna Ferrara, **2024**, University of Arizona, College of Social and Behavior Sciences, Anthropology and History, undergraduate research assistant

Yoni Goodman, **2025–2026**, University of Arizona, College of Social and Behavior Sciences, Geography and Chinese Language, undergraduate research assistant

Madeleine Zeunert, **2026**, University of Arizona, College of Science, Geosciences and Hydrology and Atmospheric Sciences, undergraduate research assistant

AJ Vasquez, **2026–present**, University of Arizona, College of Social and Behavior Sciences, Anthropology, undergraduate research assistant

Eduardo Bastian, **2026–present**, University of Arizona, College of Science, Geosciences, undergraduate research assistant

Teaching

University of Arizona — Tucson, AZ

School of Geography, Development, and Environment

GEOG 170A1 *Introduction to Physical Geography* (Spring)
GEOG 230 *Our Changing Climate* (Autumn)
GEOG 430/530 *The Climate System* (Spring)
GEOS 599-010 *Models of the North Atlantic jet (Independent Study)* (Spring 2018)
GEOG 696C *Spatiotemporal Data Analysis* (Autumn 2016, 2019, 2021, 2023, Spring 2026)
GEOG 696C *State-of-the-Art in Climate Science* (Autumn 2020)
GEOG 696F *Advanced Methods and Techniques* (Autumn 2024)
GEOG 696M *Hydroclimate: Past, Present, and Future* (Autumn 2017)
GEOG 699-013 *Topics in Paleobiogeography (Independent Study)* (Spring 2016)
GEOG 699-013 *Statistics for Biogeography (Independent Study)* (Spring 2017)

MIT-WHOI Joint Program in Oceanography — Cambridge and Woods Hole, MA

Marine Geology and Geophysics

12.708 *Topics in Paleoceanography* (Autumn 2013)

Columbia University — New York, NY

Masters in Sustainability Management Program

SUMA K4230 *The Earth's Climate System* (Spring 2011, Spring 2012)

Department of Earth and Environmental Science
EESC V2011 *Earth's Environmental Systems: The Climate System* (Spring 2012)

Grants

Current and Active | External Funding as PI or coPI: $12,185,175 since 2007

- "Renewed support for paleoCAMP: A multidisciplinary summer school for graduate students in paleoclimatology", PI: Jessica Tierney, co-PIs: **K.J. Anchukaitis**, Sylvia Dee, Daniel Ibarra, Kathleen Johnson, Heising-Simons Foundation, 01/2025 – 12/2027, $450,000
- "The North American Temperature Atlas–A Climate Field Reconstruction for Investigating Effects of Temperature on Past Droughts", PI: Karen King, coPI: **K.J. Anchukaitis**, J. Maxwell, G. Harley, E.R. Cook, NSF P4CLIMATE, AGS-2402386, 08/2024-08/2027, $34,522 (UA scope only)
- "Intertropical Convergence Zone Variations from Stable Oxygen Isotope Tree-ring Records in the Tropical Americas", PI: Laia Andreu-Hayles, coPI: **K.J. Anchukaitis** and Diego Pons, NSF P4CLIMATE, AGS-2303525, 02/2024-01/2027, $192,386 (UA scope only)
- "Indigenous forest management in a non-stationary climate", PI: Diego Pons, co-PIs: **K.J. Anchukaitis**, M. Taylor, R. DeRose, N. vonHedemann, R. Trosper, NSF Dynamics of Integrated Socio-Environmental Systems (DISES), 01/2023-06/2027, $1,599,397
- "MRI2: Acquisition of a Mini Carbon Dating System for Multi-Millennial, Multi-User Sequences of Annually Resolved 14C and Other High Precision Applications", PI: C. Pearson, co-PIs: B. Black, **K.J. Anchukaitis**, and K. Thirumalai, NSF Major Research Instrumentation Program (MRI), 09/2022-08/2025, $2,032,379
- "Tracking Divergent Warming and Tree Growth at Arctic Treeline", PI: Rosanne D'Arrigo, co-PIs: **K.J. Anchukaitis** and Benjamin Gaglioti, NSF Arctic Natural Sciences, ANS-2124889, 03/2022-02/2025, $203,991
- "High-Resolution Reconstruction of Last Millennium North American Arctic Temperatures Using Quantitative Wood Anatomy", PI: **K.J. Anchukaitis**, coPIs: Adam Csank and Stephanie McAfee, NSF Paleoclimate Perspectives on Climate Change (P2C2), AGS-2102993, 08/2021-07/2025, $471,704

Completed

- "Rainfall variability, extreme events, and vulnerability in heterogeneous social and environmental systems", PI: **K.J. Anchukaitis**, coPIs: Matthew Taylor, Diego Pons, Tom Evans, Diana Liverman, NSF Human-Environment and Geographical Sciences (HEGS), BCS-2049657, 06/2021 – 11/2025, No-Cost Extension to 04/2026, $396,593
- "PaleoCAMP (Paleoclimate training in Climate Archives, Models, and Proxies): A multidisciplinary summer school for graduate students in paleoclimatology", PI: Jessica Tierney, co-PIs: **K.J. Anchukaitis**, Tripti Bhattacharya, Daniel Ibarra, Kathleen Johnson, Heising-Simons Foundation, 01/2021 – 12/2024, $425,166
- "Pilot dendrochronology studies in the Maya Biosphere Reserve", PI: **K.J. Anchukaitis**, University of Arizona Social and Behavioral Science Research Institute (SBSRI), Faculty Small Grant, 04/2023-03/2024, $4,656
- "OpenDendro - Advanced Open-source Tools for Paleoenvironmental Reconstruction", PI: Andy Bunn, coPIs: Edward Cook, **K.J. Anchukaitis**, Tyson Swetnam, NSF Paleoclimate, AGS-2054516, 06/2021-05/2023, No-Cost Extension to 09/2024, $143,148
- "Spatiotemporal Variability in Western United States Snowpack During the Common Era", PI: **K.J. Anchukaitis**, co-PI: Bethany Coulthard, NSF Paleoclimate Perspectives on Climate Change (P2C2), AGS-1803995, 09/01/2018 – 08/31/2021, $397,055; No-Cost Extension to 08/2024

- “Drivers of Oscillations in Western US Cool-Season Precipitation”, PI: Park Williams, coPI: **K.J. Anchukaitis**. California Department of Water Resources, 10/2021 – 9/2022, No-Cost Extension to 09/2023, $213,182
- “Developing Tree-Ring Based Streamflow Reconstructions for Large and Complex River Basins”, PI: **K.J. Anchukaitis**, co-PI: Bethany Coulthard, NSF Geography and Spatial Sciences, BCS-1759629, 09/01/2018 – 02/28/2022, $279,672; No-Cost Extension to 03/2023
- “Precision dating of North American medieval megadroughts using cosmogenic radiocarbon spikes in tree-ring chronologies”, PI: **K.J. Anchukaitis**, University of Arizona Social and Behavioral Science Research Institute (SBSRI), Faculty Small Grant, 03/2022-02/2023, $4,950
- “2000 Years of Variability in the Southern Annular Mode (SAM) from Tree Rings and Ice”, PI: Amy Hessl, co-PIs: **K.J. Anchukaitis**, NSF Paleoclimate Perspectives on Climate Change (P2C2), AGS-1803946, 02/01/2019 – 01/31/2022, $369,192 (UA Scope: $149,929); No-Cost Extension to 02/2023
- “A stable isotope dendroclimatology approach toward reconstructing paleoclimate dynamics in Central America”, PIs: Laia Andreu-Hayles and **K.J. Anchukaitis**, Columbia University Climate Center, 12/2020, $8,360
- “The global climate response to volcanic eruptions in the Last Millennium Reanalysis”, PI: J. Emile-Geay, co-PIs: **K.J. Anchukaitis** and G. Hakim. NOAA Climate Program Office, Climate Monitoring Program and Climate Variability and Predictability Program, NA18OAR4310420, 09/01/2018 – 08/31/2020, $299,973 (UA Scope: $64,004)
- “Stable and Radiogenic Isotope Approach to Paleoclimate Dynamics in Central America” PI: **K.J. Anchukaitis**, University of Arizona, Research, Discovery & Innovation (RDI) International Research and Program Development Grant, 07/01/2018-06/30/2019, $26,510
- “Recent Northeastern United States Temperature Records in the Context of the Late Holocene” PIs: **K.J. Anchukaitis**, Jeff Donnelly, and Neil Pederson, NSF Paleoclimate Perspectives on Climate Change (P2C2), AGS-1304262, 08/2013 – 07/2016 (No Cost Extensions to 07/2019), $447,183
- “Past Ocean-Atmosphere Variability from Spatiotemporal Patterns of North Atlantic Climate During the Common Era” (with Lamont-Doherty Earth Observatory, PI: Ed Cook), coPIs: **K.J. Anchukaitis**, Rosanne D’Arrigo. NSF Paleoclimate Perspectives on Climate Change (P2C2), AGS-1501856, 05/2015 – 04/2018 (No Cost Extensions to 04/2019), (UA scope: $93,491)
- “Spatiotemporal Variability of Northwestern North American Temperatures in Response to Climatic Forcing” (with Lamont-Doherty Earth Observatory, PI: Rosanne D’Arrigo), coPIs: **K.J. Anchukaitis**, Laia Andreu-Hayles, Greg Wiles, Rob Wilson, NSF Paleoclimate Perspectives on Climate Change (P2C2), AGS-1501834, 05/2015 – 04/2018 (No Cost Extensions to 04/2019), (UA scope: $105,155)
- “Water Resource Relevant Hydroclimatic Reconstructions for Western North America” PI: **K.J. Anchukaitis**, Postdoctoral Researcher: Bethany Coulthard. U.S. Geological Survey (G16AC00266), 09/02/2016 – 9/01/2018, $118,886
- “Reconstructing Changes in Asian Monsoon Circulation during the Last Millennium from Stable Isotopes in Tropical Tree Rings” PIs: **K.J. Anchukaitis**, B.M. Buckley, C. Ummenhofer, NSF Paleoclimate Perspectives on Climate Change (P2C2), AGS-1203818, 8/15/2012 – 8/14/2015 (No Cost Extension to 08/2017), $688,279
- “Reconstructing Droughts in the Tropical Americas Using Tree-Ring Analysis” PIs: **K.J. Anchukaitis** and Matthew Taylor, NSF Geography and Spatial Sciences, BCS-1263609, 07/2013–12/2016, $278,599
- “CNH: Pluvials, Droughts, Energetics, and the Mongol Empire” (with West Virginia University, PI: Amy Hessl), coPIs: Neil Pederson and **K.J. Anchukaitis**, Coupled Natural-Human Systems (CNH), BCS-1210360, 05/2012 – 04/2016 (No Cost Extension to 05/2017) , $1,394,398; (Pederson & Anchukaitis scope: $317,576)

- “Tree-Ring Reconstructions of Western North Pacific Climate Dynamics”, PIs: R.D. D’Arrigo, **K.J. Anchukaitis**, N. Davi. NSF Paleoclimate Perspectives on Climate Change (P2C2) AGS-1159430, 06/2012 – 05/2015 (No Cost Extension to 05/2016), $686,982
- “Past and Future Drought Variability in the Mediterranean Basin” PI: Ramzi Touchan, coPIs: **K.J. Anchukaitis**, David Meko, and B.I. Cook, NSF Paleoclimate Perspectives on Climate Change (P2C2) AGS-1103450, 06/2011 – 05/2014, No-cost extension to 5/2015, (Anchukaitis and Cook scope: $70,388).
- “Paleoclimate shocks: environmental variability, human vulnerability, and societal adaptation during the last millennium in the Greater Mekong Basin”, PIs: B.M. Buckley, T. Heikkila, **K.J. Anchukaitis**, and 7 others, Coupled Natural-Human Systems (CNH), ATM-0908971, 10/2009 – 9/2013 (No Cost Extension to 09/2014), $1,401,351
- “Shifting Seasonality of Northern Forest Response to Arctic Environmental Change”, PIs: R.D. D’Arrigo, **K.J. Anchukaitis**, NSF ARC-0902051, 4/2009 – 3/2012, $381,154
- “Developing multicentury drought reconstructions from Guatemala and the context for past and future hydroclimatic change”, PIs: **K.J. Anchukaitis**, M. Taylor, and E. Castellanos, Geography and Spatial Sciences, NSF BCS-0852652, 04/2009 - 03/2012 (No Cost Extension to 03/2013), $203,351
- “Annual climate during the rise and fall of the Great Mongol Empire: Rapid collection of an ancient and endangered paleoclimate resource”, PIs: N. Pederson and **K.J. Anchukaitis**, Columbia University Climate Center, 11/2011, $7,850.
- “Building capacity for managing climate risk from drought in Guatemala”, PI: **K.J. Anchukaitis**, Columbia University Earth Institute Cross-Cutting Initiative, 11/2010, $24,000
- “Construction of a device for rapid cellulose extraction from wood for applications in isotope dendrochronology and organic biogeochemistry”, PI: **K.J. Anchukaitis**, Columbia University Climate Center, 05/2011, $8,000
- “MRI-R2: Acquisition of Stable Isotope Instrumentation for High Precision Paleoclimatic and Environmental Research at the Lamont Doherty Earth Observatory”, PIs: B. Linsley, B. Yan, **K.J. Anchukaitis**, P. deMenocal, P. Schlosser, NSF AGS-0959148, 5/2010 – 4/2012, $811,868
- “Dating the World Trade Center Ship”, PIs: B.M. Buckley, **K.J. Anchukaitis**, N. Pederson, E.R. Cook, R.D. D’Arrigo. AKRF, Inc., The Lower Manhattan Development Corporation, and The Port Authority of NY & NJ, 8/2010 – 7/2011, $12,306
- “Establishing the stable oxygen isotope signature of precipitation in the Central Highlands of Vietnam for application to dendroclimatology”, PI: **K.J. Anchukaitis** and B.M. Buckley, Columbia University Climate Center, 11/2008, $8,000
- “Assessing the dendrochronological potential of tree species in highland Guatemala”, PI: **K.J. Anchukaitis**, Columbia University Climate Center, 12/2007, $5,000

Honors, Awards, and Fellowships (since 2007)

Hammond Distinguished Lecturer, University of Tennessee, 2024

Willi Dansgaard Award, Paleoceanography and Paleoclimatology Section, American Geophysical Union, 2023

Udall Center for Studies in Public Policy Fellowship, University of Arizona, Autumn 2018

George H. Davis Travel Fellowship, Office of Research, Discovery & Innovation, University of Arizona, Spring 2018

Research Professorship, College of Social and Behavioral Sciences, University of Arizona, Autumn 2017

Henry Cowles Award for Excellence in Publication, Biogeography Specialty Group, Association of American Geographers, 2015

Stephen S. Visher Lecturer in Climatology, Indiana University, February 2015

Marsico Visiting Scholar, University of Denver, Autumn 2011

Newton International Postdoctoral Fellowship at Oxford University, The Royal Society (United Kingdom), 2009-2011 (*declined*)

Andrew Ellicott Douglass Memorial Scholarship, The University of Arizona, 2007

US Department of Energy (DOE) Global Change Education Program, Graduate Research and Education Fellowship (GCEP-GREF), 2004 – 2006; *declined* in 2007.

NSF Integrative Graduate Education Research and Training Fellowship (IGERT), IGERT Program for Archaeological Sciences, The University of Arizona, 2003 – 2004, 2007.

Galileo Circle Scholar, The University of Arizona College of Science, 2004

U.S. Environmental Protection Agency (EPA) STAR Fellowship, 1999 – 2001.

PRESENTATIONS Recent (previous 3 years only) † = postdoctoral author, ‡ = student author

[32] **Anchukaitis, K.J.**, S. Dee, M. Tessler, I. González-Mendez, J. Zhu, J.E. Tierney, Paleoclimate data assimilation of the Pacific Walker Circulation, Abstract PP13B-0883 presented at the 2025 Fall Meeting, American Geophysical Union, New Orleans, LA, December 15–19, 2025

[31] ‡ González-Mendez, I., **K.J. Anchukaitis**, D. Pons, K. Morino, T. Anderson, L. Andreu-Hayles, Water isotopes in tree rings track neotropical climate dynamics, Abstract PP32A-05 presented at the 2025 Fall Meeting, American Geophysical Union, New Orleans, LA, December 15–19, 2025

[30] ‡ González-Mendez, I., **K.J. Anchukaitis**, D. Pons, K. Morino, A multicentury tree-ring chronology from Abies guatemalensis in Central America, Abstract GC43A-01 presented at the 2025 Fall Meeting, American Geophysical Union, New Orleans, LA, December 15–19, 2025

[29] † Edwards, J., **K.J. Anchukaitis**, F. Molina, K. Morino, L. Andreu-Hayles, R. D'Arrigo, G. von Arx, Tree-Ring Anatomy Captures Boreal Tree Responses to Summer Heat Extremes and Atmospheric Drought, Abstract GC43P-0997 presented at the 2025 Fall Meeting, American Geophysical Union, New Orleans, LA, December 15–19, 2025

[28] King, K., E.R. Cook, **K.J. Anchukaitis**, G.L. Harley, J.T. Maxwell, A spatial field reconstruction of warm-season maximum temperatures for eastern North America: a new paleoclimate data product for evaluating temperature-hydroclimate relationships over the late Holocene, Abstract PP33C-1085 presented at the 2025 Fall Meeting, American Geophysical Union, New Orleans, LA, December 15–19, 2025

[27] † Anderson, T.G., K.A. McKinnon, I. González-Mendez, D. Pons, **K.J. Anchukaitis**, Regional variability in moisture transport and recycling in Central America, Abstract GC43A-04 presented at the 2025 Fall Meeting, American Geophysical Union, New Orleans, LA, December 15–19, 2025

[26] † Wang, F. **K.J. Anchukaitis**, E. Wise, A. Friend, B. Yang, J. Edwards, X. Jiang, M.P. Dannenberg, A New R Implementation of Process-Based Models for Simulating Tree-Ring Widths And Densities, Abstract PP33C-1083 presented at the 2025 Fall Meeting, American Geophysical Union, New Orleans, LA, December 15–19, 2025

[25] **Anchukaitis, K.J.**, The challenge of multiple proxies. Program on Climate Change Summer Institute, University of Washington, Friday Harbor, WA, September 10–12, 2025 (**Invited**)

[24] **Anchukaitis, K.J.**, J.E. Tierney, S. Dee, J. Zhu, Pacific atmospheric circulation from paleoclimate data assimilation via the ensemble Kalman filter, Poster presented at the 2025 Annual Meeting of the American Meteorological Society, New Orleans, LA, January 12-16, 2025

[23] Ibarra, D., J.E. Tierney, **K.J. Anchukaitis**, T. Bhattacharya, K. Johnson, paleoCAMP (Paleoclimate Training in Climate Archives, Models, and Proxies): Developing an Inclusive Graduate Student Summer School in Paleoclimatology, Abstract ED51A-04 presented at the 2024 Fall Meeting, American Geophysical Union, Washington, DC, December 13-17, 2024 (**Invited**)

[22] ‡ Harris, K., J.E. Tierney, S. Muñoz, **K.J. Anchukaitis**, Decadal-Scale Variability in the North American Monsoon system during the Common Era, Abstract PP42B-02 presented at the 2024 Fall Meeting, American Geophysical Union, Washington, DC, December 13-17, 2024

[21] ‡ González-Mendez, I., **K.J. Anchukaitis**, C. Pearson, D. Pons, D. Frank, J. Edwards, E. Luna, W. Pérez, Annual chronology and climate signals in *Swietenia macrophylla* and *Cedrela odorata* (Meliaceae) in the Maya Lowlands, Abstract PP52A-03 presented at the 2024 Fall Meeting, American Geophysical Union, Washington, DC, December 13-17, 2024

[20] Williams, A.P., K.A. McKinnon, **K.J. Anchukaitis**, A. Gershunov, A.M. Varuolo-Clarke, R.E.S Clemesha, H. Liu, Anthropogenic intensification of cool-season precipitation is not yet detectable across the western United States, Abstract A23G-2054 presented at the 2024 Fall Meeting, American Geophysical Union, Washington, DC, December 13-17, 2024

[19] Cook, B.I., E.R. Cook, **K.J. Anchukaitis**, D. Singh, Characterizing the 2010 Russian heatwave-Pakistan flood concurrent extreme over the last millennium using the Great Eurasian Drought Atlas, Abstract PP52A-01 presented at the 2024 Fall Meeting, American Geophysical Union, Washington, DC, December 13-17, 2024

[18] ‡ Anderson, T.G., D. Pons, M.J. Taylor, A. Xuruc, H. Rodríguez-Salvatierra, Z. Guido, J.A. Sullivan, D. Liverman, **K.J. Anchukaitis**, Complexity and mediating factors in farmers' climate perceptions and agricultural adaptation strategies in the Guatemalan Dry Corridor, Paper presented at the 38th Conference of Latin American Geography, Old San Juan, Puerto Rico, May 22-26, 2024

[17] **Anchukaitis, K.J.**, J.A. Edwards, E.R Cook, J.R. Southon, B. Gaglioti, B. Hatchett, P. Krusic, A.P. Williams, K. King, S.W. Stine, Chronology and climate context for medieval megadrought in the eastern Sierra Nevada, Paper presented at the annual meeting of the American Association of Geographers, Honolulu, Hawaii, April 16-20, 2024

[16] Leland, C., N. Davi, R.D. D'Arrigo, L. Andreu-Hayles, A. Pacheco-Solana, T. Porter, M. Pisaric, **K.J. Anchukaitis**, J. Edwards, S. Beaulieu The Curious Case of 1959: Tree-ring Evidence of an Extreme Cold Event in Northwestern North America, Paper presented at the annual meeting of the American Association of Geographers, Honolulu, Hawaii, April 16-20, 2024

[15] **Anchukaitis, K.J.**, All constraints but nature's: hydroclimate in western North America during the Common Era, Hammond Distinguished Lecture, University of Tennessee, April 4, 2024 (**Invited**)

[14] **Anchukaitis, K.J.**, Z.K. Sethuraju, D. Pons, J. Edwards, T.G. Anderson, I. Gonzalez-Mendez, M. Taylor, Dendrochronological evidence for the impacts of the 1902 Santa Maria volcanic eruption on the forests of western Guatemala, 12th Cities on Volcanoes conference, Antigua, Guatemala, February 11-17, 2024

[13] ‡ Anderson, T.G., K. McKinnon, D. Pons, **K.J. Anchukaitis**, How exceptional was the 2015-2019 Central American Drought? Abstract A43H-03 presented at the 2023 Fall Meeting, American Geophysical Union, San Francisco, CA, December 10-16, 2023

[12] **Anchukaitis, K.J.**, J.A. Edwards, E.R Cook, J.R. Southon, B. Gaglioti, B. Hatchett, P. Krusic, A.P. Williams, K. King, S.W. Stine, Chronology and climate context for medieval megadrought in the eastern Sierra Nevada, Abstract PP32B-02 presented at the 2023 Fall Meeting, American Geophysical Union, San Francisco, CA, December 10-16, 2023

[11] King, K.E., E.R. Cook, **K.J. Anchukaitis**, B.I. Cook, J.E. Smerdon, R. Seager, G. Harley, B. Spei, The Increasing Prevalence of Hot Drought Across Western North America Since the

16th Century, Abstract PP32B-03 presented at the 2023 Fall Meeting, American Geophysical Union, San Francisco, CA, December 10-16, 2023

[10] ‡ Edwards, J., **K.J. Anchukaitis**, S. McAfee, A. Csank, L. Andreu-Hayles, R. D'Arrigo, G. von Arx, Reconstructing Past Arctic Climate Variability through High-Resolution Tree-Ring Wood Anatomical Chronologies from Northern Alaska, Abstract PP32B-04 presented at the 2023 Fall Meeting, American Geophysical Union, San Francisco, CA, December 10-16, 2023

[9] **Anchukaitis, K.J.**, Reconstructed snowpack variability in the western United States over the last millennium, ESS Seminar Series, Stanford Doerr School of Sustainability, Stanford University, May 11, 2023 (**Invited**)

[8] ‡ Edwards, J., **K.J. Anchukaitis**, S. McAfee, A. Csank, L. Andreu-Hayles, R. D'Arrigo, G. von Arx, Records of past temperatures in wood anatomy of high-latitude trees, Paper presented at TRACE 2023 - Tree-Rings in Archaeology, Climatology and Ecology, Coimbra, Portugal, May 8-13, 2023

[7] Burke, A., H. Innes, L. Crick, **K.J. Anchukaitis**, W. Hutchison, J. McConnell, J. Rae, M. Sigl, R. Wilson, High-resolution sulfur isotopes from ice cores: improved estimates of the volcanic forcing of climate, EGU General Assembly 2023, EGU23-7109, https://doi.org/10.5194/egusphere-egu23-7109, Vienna, Austria, April 24–28, 2023.

[6] ‡ González-Mendez, A.I., **K.J. Anchukaitis**, Z. Sethuraju, T.G. Anderson, K. Morino, D. Pons, M. Taylor, Dendrochronology and quantitative wood anatomy of Guatemala's high elevation conifers, Paper presented at the annual meeting of the American Association of Geographers, Denver, Colorado, March 23-27, 2023

[5] ‡ Anderson, T.G., K. McKinnon, D. Pons, **K.J. Anchukaitis**, How exceptional was the 2015–2019 Central American Drought? Paper presented at the annual meeting of the American Association of Geographers, Denver, Colorado, March 23-27, 2023

[4] ‡ Edwards, J., **K.J. Anchukaitis**, S. McAfee, A. Csank, L. Andreu-Hayles, R. D'Arrigo, G. von Arx, Records of past temperatures in wood anatomy of high-latitude trees, Paper presented at the annual meeting of the American Association of Geographers, Denver, Colorado, March 23-27, 2023

[3] ‡ Harris, K., J.E. Tierney, S. Munoz, **K.J. Anchukaitis**, Late Holocene sea surface temperatures and monsoon rainfall variability in the Gulf of California, Paper presented at the annual meeting of the American Association of Geographers, Denver, Colorado, March 23-27, 2023

[2] **Anchukaitis, K.J.**, B. Coulthard, N. Gauthier, G.T. Pederson, S.T. Gray, Reconstructed snowpack variability in the western United States over the last millennium, Paper presented at the annual meeting of the American Association of Geographers, Denver, Colorado, March 23-27, 2023

[1] **Anchukaitis, K.J.**, Reconstructed snowpack variability in the western United States over the last millennium, The Glenn and Susan Buckley Lecture in Environmental Geology, University of Illinois Urbana-Champaign, February 9, 2023 (**Invited**)

Service & Synergistic Activities

University of Arizona, 2015–: College of Social and Behavioral Sciences Promotion and Tenure Committee (2021); School of Geography, Development, and Environment Promotion and Tenure Committees (2021, 2023, 2024); School of Geography, Development, and Environment Colloquium Committee (2017 – 2018; 2019 – 2021); Chair, Global Change Graduate Interdisciplinary Program (2017 – 2025); School of Geography, Development, and Environment Annual Performance Review Committee (2018 – 2021; 2024 – 2025; Chair in 2020–2021); School of Geography, Development, and Environment Committee for Diversity, Equity, and Inclusion (2019–2020); Graduate Admissions Committee, School of Geography and Development (2015 to 2018; 2023 – 2025); Earth System Genomics faculty search committee

(2016); Integrated Land Use Science/Human-Environment Systems faculty search committee (2016–2017); Forests, Carbon & Climate faculty search committee (2017); Paleoclimatology Faculty Search committee (2017–2018); Data Science and Spatial Analysis Faculty Search committee (2019–2020); School of Geography, Development, and Environment Director Search committee (2023–2025); Physical Geography Faculty Search committee (2025–2026)

Organizer, co-PI, and instructor, paleoCAMP: Paleoclimate Training in Climate Archives, Models, and Proxies, 2022 – present

Member, National Academies of Sciences, Engineering, and Medicine, Committee on Functions and Criteria for a New Center for Paleoenvironmental Records of Extreme Events, 2025 – 2026

Associate Editor, *Dendrochronologia*, 2023 – 2026

Editorial Advisory Board, *Quaternary Science Reviews*, 2008 – 2025.

President-elect, The American Quaternary Association (AMQUA), 2022 – 2024

Steering Group and co-founder, PAGES Working Group on Volcanic Impacts on Climate and Society 2015 – 2025

Guest Instructor, 'Climate Diplomacy', School of Professional and Area Studies (SPAS), Foreign Service Institute (FSI), 2021 – 2023

Guest editor, special issue on 'Interdisciplinary studies of volcanic impacts on climate and society' for *Climate of the Past*, 2020 – 2024

Member, PAGES Asia2k (2011–2014), Ocean2k (2011 – 2015), North America 2k (2015 – present) Working Groups.

U.S. National Committee (USNC) for the International Union for Quaternary Research, National Academy of Sciences, 2018 – 2023

Program Committee chair, Conference of Latin American Geography (CLAG) meeting, Tucson, AZ, 2021 – 2023

Contributing Author, Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report (AR6), 2019 – 2021

Organizer and co-chair, 'Climate of the Common Era' for the American Geophysical Union Annual Meeting (AGU), 2010 – 2019; 'Monsoon Effects on Past Civilizations', 2012.

US Secretariat and Scientific Committee, 10th International Conference on Dendrochronology, Thimphu, Bhutan (2018)

University of Arizona Museum of Art panelist, 'Fires of Change' exhibit (2016).

Coordinator, PAGES2k Transregional Reconstruction Methods Development, 2014 – 2016.

Scientific Committee, 3rd American Dendrochronology Conference, Mendoza, Argentina, March 2016

Guest editor, *PAGES Magazine*, 2015.

Founder, organizer, and lead instructor, Central American Dendroecological Fieldweek, Chancol, Cuchumatanes, Guatemala, March 2015.

Fellow, Expert Witness Training Academy, NSF Paleoclimate Program and William Mitchell College of Law, St. Paul, Minnesota, August 2015.

Chair and co-organizer, PAGES Advances in Climate Field Reconstructions, April 15-16, 2014, Woods Hole, Massachusetts.

Scientific Advisory Committee, 9th International Conference on Dendrochronology, Melbourne, Australia, January 2014.

Contributing Author, Intergovernmental Panel on Climate Change (IPCC) Fifth Assessment Report (AR5; 2013)

Instructor, North American Dendroecological Fieldweek, Black Rock Forest, New York, June-July 2013.

Arctic Research Consortium of the U.S. (ARCUS) PolarTREC Selection Committee, 2012–2013

International Workshop on Climate Informatics, Program Committee and Breakout Group Leader, 2012

Lamont-Doherty Earth Observatory, 2007–2012: Columbia Climate Center, 2011 - 2012; Promotion and Careers Committee, 2009 - 2012; Campus Life Committee, 2008 - 2010; Volunteer and Speaker, Annual Open House, 2007, 2008, Panelist on Climate Extremes, 2011.

Organizer and co-chair, 'Advances in Paleoclimatology' for the Annual Meetings of the Association of American Geographers, 2004 – 2009; 'Challenges in Dendrochronology', 2012.

Mentor, 'Google Summer of Code', Climate Code Foundation, Summer 2011.

Guest editor, *Dendrochronologia*, 2003 – 2004.

Professional reviewer of journal manuscripts for *Science*, *Nature*, *Nature Geoscience*, *Nature Climate Change*, *Science Advances*, *Proceedings of the US National Academy of Sciences*, *Journal of Climate*, *Geology*, *Journal of Geophysical Research*, *Geophysical Research Letters*, *Paleoceanography and Paleoclimatology*, *Climate Dynamics*, *Physical Geography*, *Geochimica et Cosmochimica Acta*, *Ecology*, *Earth and Planetary Science Letters*, *Global Change Biology*, *Climate of the Past*, *Quaternary Research*, *Climatic Change*, *The Holocene*, *Geochemistry, Geophysics, Geosystems*, *Tellus B*, *Chemical Geology*, *Canadian Journal of Forest Research*, *Forest Ecology and Management*, *International Journal of Biometeorology*, *Canadian Water Resources Journal*, *Journal of Volcanology*, *Earth Interactions*, *Historical Geography*, *Current Anthropology*, *Tree-Ring Research*, *Dendrochronologia*, and others.

Proposal reviewer for the National Science Foundation (NSF, USA), National Aeronautics and Space Administration (NASA, USA), Natural Environment Research Council (NERC, UK), Deutsche Forschungsgemeinschaft (the German Research Foundation, DFG), the Swiss National Science Foundation, Fonds de Recherche du Québec, NAS/USAID PEER, and the National Geographic Society.